



SECP3723 - 01

SYSTEM DEVELOPMENT TECHNOLOGY

INDIVIDUAL PROJECT

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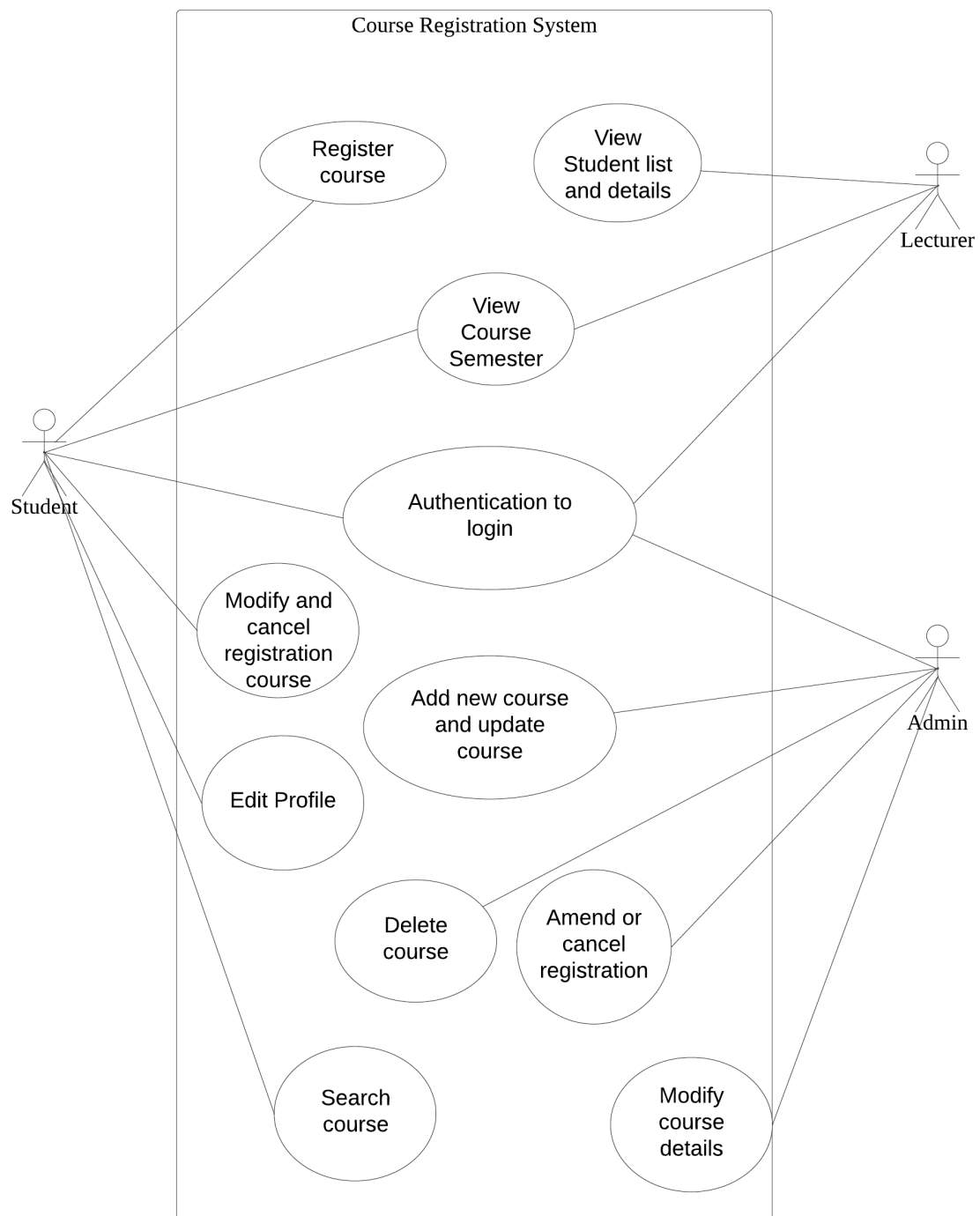
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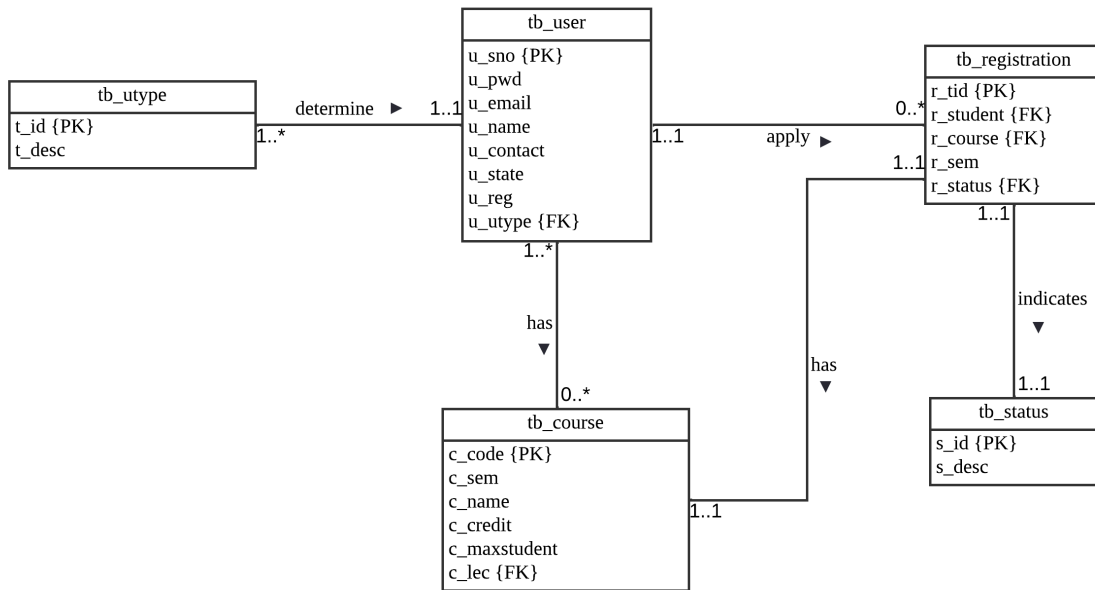
SYSTEM INTRODUCTION

Developing a Course Registration System (CRS) using XAMPP and phpMyAdmin provides an efficient way to manage courses, instructors, and students in an educational environment. XAMPP, a local development server that includes Apache, MySQL, PHP, and Perl, enables seamless web application development, while phpMyAdmin simplifies database management through a user-friendly interface. This system will allow administrators to create and organize courses, assign instructors, and manage student enrollments, while instructors can upload materials, track student progress, and interact with learners. Students will be able to enroll in courses, access learning resources, and submit assignments, creating a structured and interactive learning platform. By leveraging XAMPP and phpMyAdmin, this project ensures a practical and scalable solution for educational institutions looking to digitize their course management processes.

USE CASE



ENTITY RELATIONSHIP DIAGRAM



DEVELOPMENT SOFTWARE, LANGUAGE TECHNOLOGY AND TOOLS

Development Software

1. Xampp
2. MySQL - MyPHPAdmin

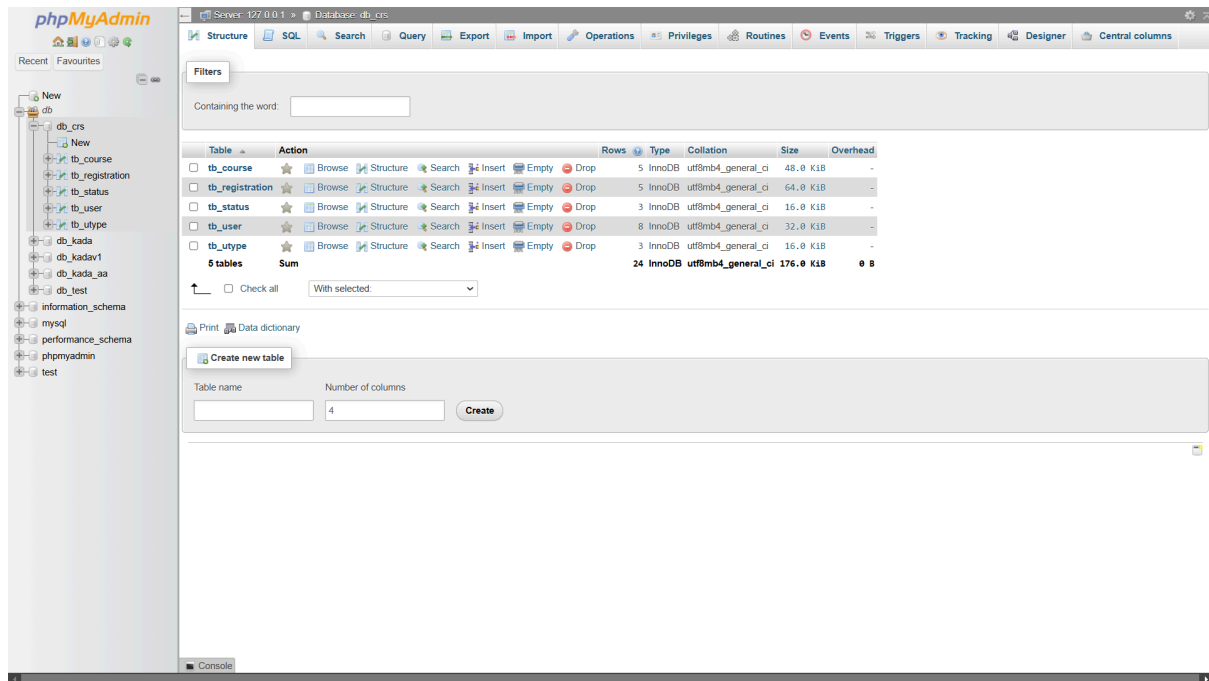
Language Technology

1. HTML
2. CSS
3. PHP
4. JAVA SCRIPT
5. BOOTSTRAP

Tools

1. VS Code
2. Lucid Chart

DATABASE DESIGN



Database db_crs

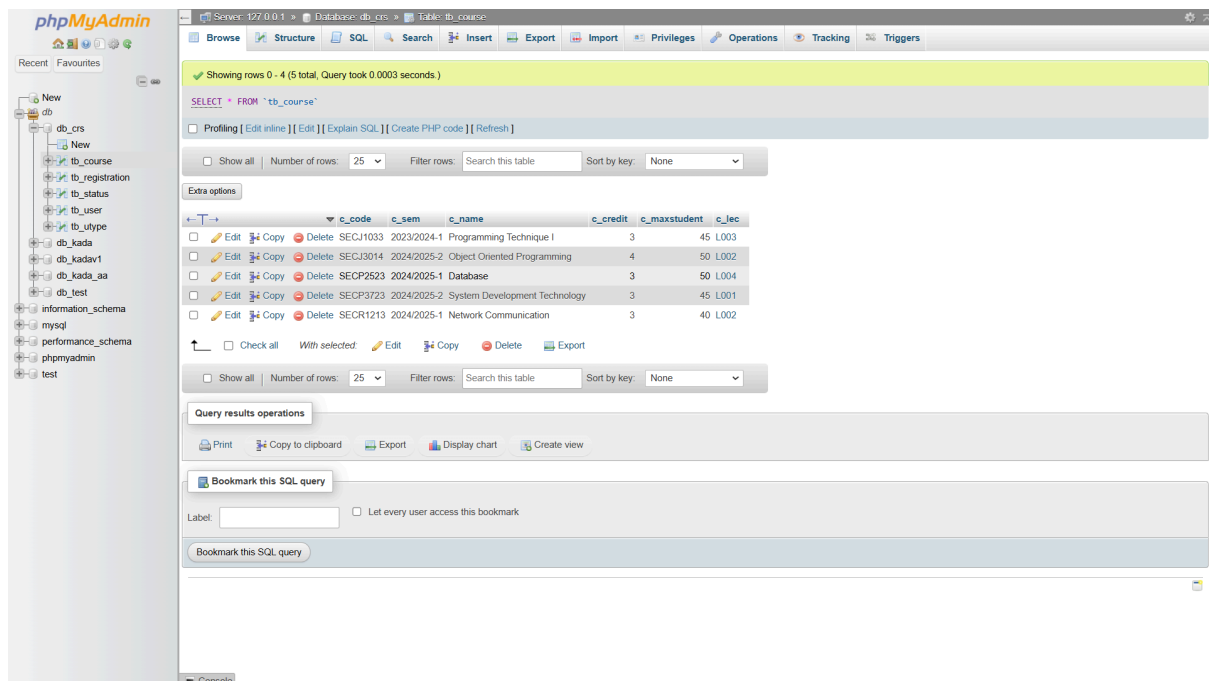


Table tb_course

phpMyAdmin

Server: 127.0.0.1 » Database: db_crs » Table: tb_registration

Showing rows 0 - 4 (5 total, Query took 0.0003 seconds)

SELECT * FROM `tb\_registration`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	r_id	r_student	r_course	r_sem	r_status
☐	12	S001	SECP3723	2024/2025-2	2
☐	15	S011	SECR1213	2024/2025-1	2
☐	17	S001	SECP2523	2024/2025-1	1
☐	18	S001	SECI1033	2023/2024-1	1
☐	19	S011	SECI1033	2023/2024-1	1

Check all | With selected: Edit | Copy | Delete | Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print | Copy to clipboard | Export | Display chart | Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

Table tb_registration

phpMyAdmin

Server: 127.0.0.1 » Database: db_crs » Table: tb_status

Showing rows 0 - 2 (3 total, Query took 0.0003 seconds)

SELECT * FROM `tb\_status`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	s_id	s_desc
☐	1	Received
☐	2	Approved
☐	3	Rejected

Check all | With selected: Edit | Copy | Delete | Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print | Copy to clipboard | Export | Display chart | Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

Table tb_status

phpMyAdmin

Server: 127.0.0.1 » Database: db_crs » Table: tb_user

Showing rows 0 - 7 (8 total, Query took 0.0003 seconds)

SELECT * FROM `tb\_user`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		u_sno	u_pwd	u_email	u_name	u_contact	u_state	u_reg	u_utype
☐	Edit Copy Delete	A001	\$2y\$10\$Bp6WP8Mps7snfOwkUkDukeVScrdkDpAhqBgnShSfr...	minul@gmail.com	Aminul Hasan	0179072912	Pulau Pinang	2025-01-31 03:17:24	3
☐	Edit Copy Delete	L001	\$2y\$10\$dUlwYOikvOdBK7HDFD02oOIU2Prc3AUghKS49k...	aina@gmail.com	Aina Abdul	01911111111	Johor	2023-02-01 00:00:00	1
☐	Edit Copy Delete	L002	\$2y\$10\$H377fbbAKPZuslmyn4u1sQvgZ4gCTezkhh6wVOv...	faz@gmail.com	Fazura Abdul	01722222222	Kelantan	2023-10-01 00:00:00	1
☐	Edit Copy Delete	L003	\$2y\$10\$ZazAunAVyKfgZun9x0ZM7uT0yysSUCIPp89pTZfjal...	ahamd@gmail.com	Ahmad Kasim	0192342351	Kelantan	2025-01-28 15:49:30	1
☐	Edit Copy Delete	L004	\$2y\$10\$P217tsNehHyOR9SStbRF-rYDIAFWJMDs0mmvzPZy9...	nur@gmail.com	Nur Fatimah	012832355	Wilayah Persekutuan	2025-01-28 15:49:30	1
☐	Edit Copy Delete	S001	\$2y\$10\$01stvA3m.wq9KdUopH.EX01jivZQvzF3LywkwKW...	fai@gmail.com	Fatah Abdul	0113971310	Melaka	2024-02-01 00:00:00	2
☐	Edit Copy Delete	S002	\$2y\$10\$NEi8LheyJa.jy3vDTXcRse7vPeif3dP7ToGhPWbdQ...	kam@gmail.com	Kamarul Abdul	0124444444	Sarawak	2024-10-01 00:00:00	2
☐	Edit Copy Delete	S011	\$2y\$10\$MZLdV32d8PZyooM7CCPSuytWfJdJLcFxpJBLPovNp...	afiq@gmail.com	Aliq Daniai	0194312944	Johor	2024-11-19 23:14:44	2

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

Table tb_user

phpMyAdmin

Server: 127.0.0.1 » Database: db_crs » Table: tb_utype

Showing rows 0 - 2 (3 total, Query took 0.0002 seconds)

SELECT * FROM `tb\_utype`

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

		t_id	t_desc
☐	Edit Copy Delete	1	Lecturer
☐	Edit Copy Delete	2	Student
☐	Edit Copy Delete	3	Admin

Check all | With selected: Edit Copy Delete Export

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations

Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Label: ☐ Let every user access this bookmark

Bookmark this SQL query

Console

Table tb_utype

DEVELOPMENT STEPS

1. Planning and Requirements Gathering

- Define the objectives of the CRS (e.g., course management, student enrollment, edit profile, registration management).
- Identify the user roles: Admin, Lecturer, and Student.

2. Setting Up the Development Environment

- Install XAMPP to create a local server with Apache, PHP, and MySQL.
- Use phpMyAdmin to manage the database and create necessary tables.
- Set up a code editor like VS Code or Sublime Text for PHP development.

3. Database Design and Implementation

- Create a MySQL database (db_crs) using phpMyAdmin.
- Design tables for Users (Admin, Lecturer, Student), Courses, User type, and Status.
- Define primary keys, foreign keys, and relationships between tables.

4. Backend Development (PHP & MySQL)

- Develop the user authentication system (login, registration, roles).
- Implement CRUD (Create, Read, Update, Delete) operations for courses and users.

5. Frontend Development (HTML, CSS, JavaScript)

- Design a user-friendly interface using HTML, CSS (Bootstrap for responsiveness).
- Implement JavaScript (or jQuery) for interactive elements (e.g., form validation, AJAX requests).

6. Testing and Debugging

- Test user authentication and role-based access controls.
- Ensure proper data handling (course management, student enrollment, edit profile, registration management).
- Debug errors and optimize performance.

7. Deployment and Maintenance

- Deploy the CRS on a live server using a web hosting service with PHP and MySQL support.
- Regularly update the system based on user feedback and security improvements.

SYSTEM INTERFACE FOR ALL USERS

STUDENT INTERFACE

Course Registration System

HomeRegisterLogin

Course Management System

Register

Create a new account to access courses.

Register Now

Login

Access your account and enrolled courses.

Login

My Profile

Update your personal information and settings.

Edit Profile

Announcements

New course "Introduction to AI" now available!

System maintenance scheduled for this weekend.

Don't forget to submit your assignments before the deadline.

Quick Links

[Academic Calendar](#)

[Student Resources](#)

[Help & Support](#)

CRS developed by Afliq @ 2024

First page of CRS

Course Registration System

HomeRegisterLogin

Please Fill All Following Details

Please Enter Your Staff or Student Number

Enter your staff or student ID

Create Your Password

Enter your password

Confirm Your Password

Confirm your password

Enter Your Email Address

Enter your email address

Enter Your Full Name

Enter your full name

Enter Your Contact Number

Enter your contact number

Select Your State

Johor

Reset

Submit

CRS developed by Afliq @ 2024

Register CRS system

Please Fill In The Login

Please Enter Your Staff or Student Number

Enter your staff or student ID

Password

Enter your password

Login

Authenticate Login

Welcome, Fatah Abdul

My Courses

View your enrolled courses and course materials.

Go to Courses

My Grades

Check your grades and academic progress.

View Grades

My Profile

Update your personal information and settings.

Edit Profile

Student Homepage

Course Registration Form

Available Courses

Showing 1 to 5 of 5 entries

Search:

Course Code	Course Name	Credit Hours	Number of Student	Lecturer	Action
SECJ1033	Programming Technique I	3	2/45	Ahmad Kasim	Select
SECJ3014	Object Oriented Programming	4	0/50	Fazura Abdul	Select
SECP2523	Database	3	1/50	Nur Fatimah	Select
SECP3723	System Development Technology	3	1/45	Aina Abdul	Select
SECR1213	Network Communication	3	1/40	Fazura Abdul	Select

Previous1Next

Selected Course

Register Course

Register Course

CRS StudentHomeRegister CourseView Your CoursesSettings

Semester: 2023/2024-1

Transaction ID	Course	Course Name	Status	Modify
18	SECJ1033	Programming Technique I	Received	Modify

Semester: 2023/2024-2

Transaction ID	Course	Course Name	Status	Modify

Semester: 2024/2025-1

Transaction ID	Course	Course Name	Status	Modify
17	SECP2523	Database	Received	Modify

CRS developed by Afiq @ 2024

View Registered Course

Course Details

Course Information

Course Code: SEC1033

Course Name: Programming Technique I

Course Credit (Hour): 3

Lecturer In Charge: Ahmad Kasim

← Back to Course List

Course Actions

Cancel Course

CRS developed by Aliq @ 2024

Cancel Registered Course

Student Profile

Fatah Abdul

Student Number: S001

Email Address:

fat@gmail.com

Password:

.....

Contact Number:

0113971310

State:

Melaka

Registration Date:

2024-02-01 00:00:00

Update Profile

CRS developed by Aliq @ 2024

Edit Student Profile

LECTURER INTERFACE

CRS Lecturer

HomeView Assign Course

Settings

Welcome, Aina Abdul

View The Assigned Courses

View student enrolled courses and course materials.

Go to Courses

CRS developed by Aliq @ 2024

Lecturer Homepage

CRS Lecturer

HomeView Assign Course

Settings

Semester: 2023/2024-1

Course	Course Name	Credit Hour	Lecturer	Details
SECI1033	Programming Technique I	3	Ahmad Kasim	Detail

Semester: 2023/2024-2

Course	Course Name	Credit Hour	Lecturer	Details
No courses available for this semester.				

Semester: 2024/2025-1

Course	Course Name	Credit Hour	Lecturer	Details
SECP2523	Database	3	Nur Fatimah	Detail
SECR1213	Network Communication	3	Fazura Abdul	Detail

Semester: 2024/2025-2

CRS developed by Aliq @ 2024

View Assigned Course

Course Details

Course Information

Course Code: SEC1033

Course Name: Programming Technique I

Course Credit (Hour): 3

Semester: 2023/2024-1

Lecturer In Charge: Ahmad Kasim

Back to Assign Course List

Enrolled Students

S001 - Fatah AbdulDetails

S011 - Afiq DanialDetails

View Course Details and Student List

Course Details

Student Information

Student Number: S011

Name: Afiq Danial

Email Address: afiq@gmail.com

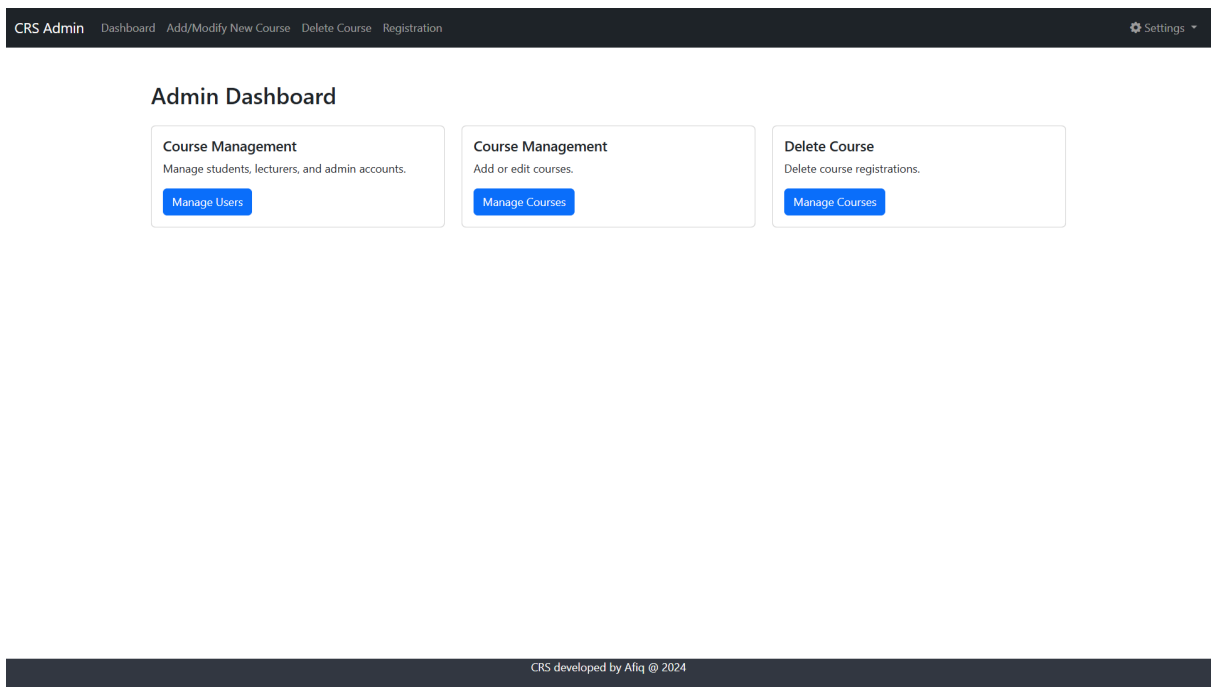
State Address: Johor

Registration Time: 2024-11-19 23:14:44

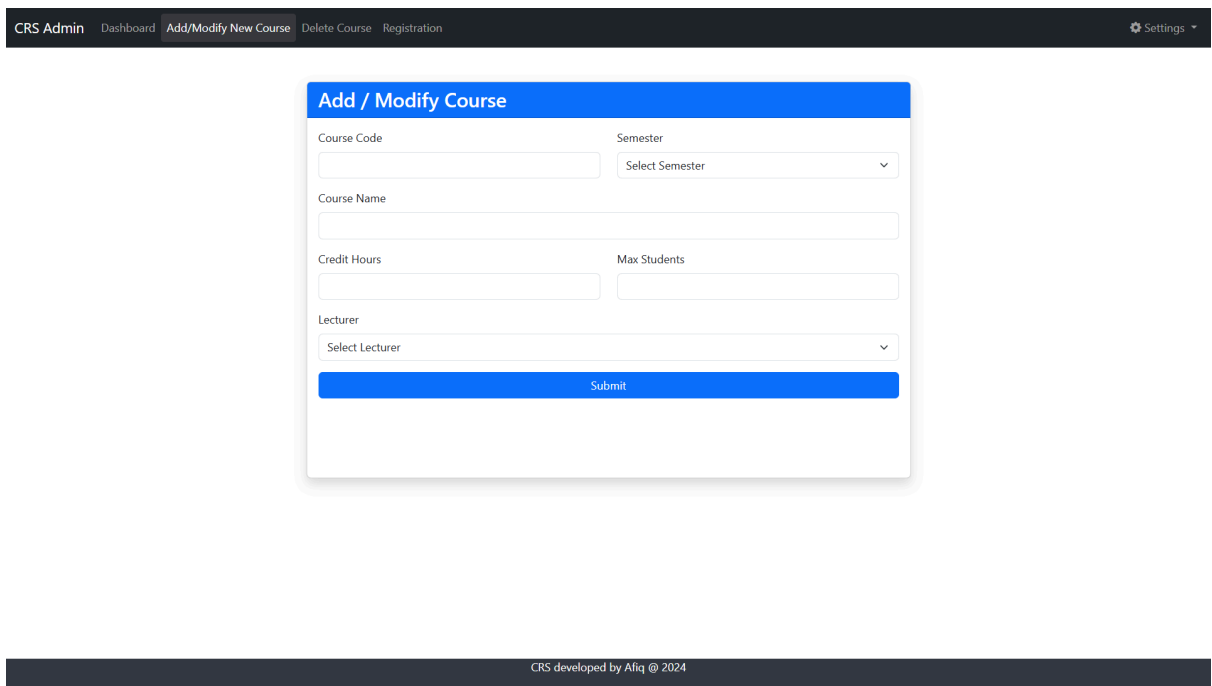
Back

View Student Details

ADMIN INTERFACE



Admin Dashboard



Add or Update Course

Course Details

Course Code	Course Name	Semester	Credit Hours	Maximum Student	Lecturer	Operation
SECJ1033	Programming Technique I	2023/2024-1	3	45	L003 - Ahmad Kasim	Delete
SECJ3014	Object Oriented Programming	2024/2025-2	4	50	L002 - Fazura Abdul	Delete
SECP2523	Database	2024/2025-1	3	50	L004 - Nur Fatimah	Delete
SECP3723	System Development Technology	2024/2025-2	3	45	L001 - Aina Abdul	Delete
SECR1213	Network Communication	2024/2025-1	3	40	L002 - Fazura Abdul	Delete

Delete Course

Transaction ID	Semester	Student ID	Student Name	Course	Course Name	Status	Operation
17	2024/2025-1	S001	Fatah Abdul	SECP2523	Database	Received	Approval
18	2023/2024-1	S001	Fatah Abdul	SECJ1033	Programming Technique I	Received	Approval
19	2023/2024-1	S011	Afiq Danial	SECJ1033	Programming Technique I	Received	Approval

List Student Registration Course

Course Approval Process

Student Information

Transaction ID: 17

Semester: 2024/2025-1

Student ID: S001

Student Name: Fatah Abdul

Course Information

Course Code: SECP2523

Course Name: Database

Current Status:

Approval Action

Approve

Reject

Back to Course Approval List

Approve or Reject Student Registration

LOCALHOST SETUP

Setting up a localhost using XAMPP is a straightforward process that allows developers to run web applications locally before deploying them online. First, download and install XAMPP from the official Apache Friends website, ensuring that the necessary components like Apache, MySQL, PHP, and phpMyAdmin are included. After installation, open the XAMPP Control Panel and start the Apache and MySQL modules to enable the web server and database management. Once running, you can access the localhost by opening a web browser and entering `http://localhost/` in the address bar. To manage databases, navigate to `http://localhost/phpmyadmin/`, where you can create and modify MySQL databases using a graphical interface. Web projects should be placed in the `htdocs` folder inside the XAMPP installation directory (e.g., `C:\xampp\htdocs\` on Windows). By using this setup, developers can test PHP applications, connect to MySQL databases, and ensure everything functions correctly in a controlled local environment before deployment.

USER CREDENTIAL

Admin

Username: A001

Password: admin123

Lecturer

Username: L001/L002/L003/L004

Password: 123456

Student

Username: S001/S002

Password: 123456